

Module 3. Data Preprocessing

- ❑ Data Quality, Data Cleaning and Data Integration
- ❑ Data Transformation
- ❑ Dimensionality Reduction
- ❑ Summary

What is Data Preprocessing? — Major Tasks

☐ Data cleaning

- ☐ Handle missing data, smooth noisy data, identify or remove outliers, and resolve inconsistencies

☐ Data integration

- ☐ Integration of multiple databases, data cubes, or files

☐ Data reduction

- ☐ Dimensionality reduction
- ☐ Numerosity reduction
- ☐ Data compression

☐ Data transformation and data discretization

- ☐ Normalization
- ☐ Concept hierarchy generation

Why Preprocess the Data? — Data Quality Issues

- ❑ Measures for data quality: A multidimensional view
 - ❑ Accuracy: correct or wrong, accurate or not
 - ❑ Completeness: not recorded, unavailable, ...
 - ❑ Consistency: some modified but some not, dangling, ...
 - ❑ Timeliness: timely update?
 - ❑ Believability: how trustable the data are correct?
 - ❑ Interpretability: how easily the data can be understood?